Smooth Graph Travel-Time Fields on SE(3):
Implementation, Methodology and Results

Implementation of *SE3_Smooth_Graph_Implementation.pdf* (v1.0)
in `G-Shape-NTFields/models/graph_method`

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Abstract

We implement the smooth-graph formulation of pairwise travel-time fields on SE(3): a network predicts a smooth latent $Z_\theta(x, y) = (a, b)$ for a start/goal pose pair and an explicit analytic decoder $T = \sqrt{a/s_t^2 + \phi(b)/s_r^2}$ reconstructs the possibly non-smooth travel time, trained with intrinsic endpoint Eikonal and finite-step Bellman constraints. The network follows the G-Shape-NTFields encoder. On the obstacle-free benchmark of the specification (stages A–D, 3 seeds, held-out test sets) the proposed `graph_gram` head reaches a median relative time error of $0.105 \pm 0.003\,\%$ (95th percentile $0.31\,\%$), reproduces the rotational cut-locus corner (exact half-turn error 0.002) and plans with $100\,\%$ success at cost ratio 1.000, whereas the G-Shape metric head trained with identical data, budget and planner reaches $9.5\,\%$ / $23\,\%$ and $98.6\,\%$ success at cost ratio 1.12. The decisive ingredients were the Fourier frequency for 12-dimensional pose features, a positive-semidefinite Gram rotation latent, a linear input skip, a relative-angle curriculum and cosine learning-rate decay. We identify a projection-metric degeneracy of the endpoint Eikonal equation that limits training without labelled anchors. In an obstacle extension (stage E) with the unmodified Shape-NTFields planner, a graph head whose translation-type latent is a metric-head distance is on par with the metric head inside the same pipeline across three environments ($81.2\,\%$ vs. $81.0\,\%$ mean success), while the Euclidean-latent `graph_gram` lags; the native NTFields training pipeline remains ahead (by 1–3 points on the T-shape benchmark, by more on the other two environments according to its own records).

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1 Introduction

Neural travel-time fields (NTFields) learn the minimum travel time $T(x, y)$ between configurations x and y by enforcing an Eikonal equation; following $-\nabla T$ yields a path. The G-Shape-NTFields baseline represents T as a learned metric between endpoint embeddings. For a rigid body the configuration space is SE(3) and the travel time has two singular sets: the diagonal $x = y$, where T behaves like a norm (a cone), and the rotational cut locus (relative rotation angle π), where two shortest rotations tie and T has a ridge. Smooth networks have to round off such corners.

The specification proposes to represent instead a *smooth* latent quantity and to recover the non-smooth time with an explicit decoder. Its central hypothesis is that *a smooth latent with an explicit non-smooth decoder approximates the pairwise travel-time field more accurately near the rotational cut locus than a directly smooth network.* The obstacle-free problem has a known analytic solution, so learning it validates the mechanism rather than solving a new problem.

Scope. We implement the complete specification (geometry, oracle, decoder, interface, losses, data, planners, training, evaluation, tests), compare the proposed field with the G-Shape metric head, a direct scalar network and further heads under matched conditions, document every experiment, and test a separate obstacle extension on the 3D-shape datasets of the repository.

Outcomes.

- Stage A: the analytic construction is verified to machine precision (19 unit tests).

- Stages B–D: the proposed head is about 90× more accurate than the metric head and plans optimally; it meets the provisional gates of the specification (median $< 1\,\%$, 95th percentile $< 5\,\%$, $\geq 99\,\%$ planning success) on all seeds (Section 6).
- The development history identifies four non-obvious failure modes and their fixes (Section 5).
- Stage E: with a metric-type translation latent the graph field is on par with the metric head in the same pipeline across three environments (Section 7).

2 Problem formulation

2.1 Poses, embedding and motion model

A pose is $x = (t_x, R_x)$ with $t_x \in \mathbb{R}^3$ and $R_x \in \mathrm{SO}(3)$; composition is $x \cdot y = (t_x + R_x t_y,\ R_x R_y)$. The fixed smooth embedding used everywhere is

$$G(x) = \big(t_x,\ \mathrm{vec}\, R_x\big) \in \mathbb{R}^{12}. \tag{1}$$

The body moves with a body-frame twist $\xi = (v, \omega)$, $\dot{x} = x\,\hat{\xi}$, under the ellipsoidal speed limit

$$\frac{\|v\|^2}{s_t^2} + \frac{\|\omega\|^2}{s_r^2} \leq 1, \qquad \text{equivalently} \qquad \xi = Su, \quad \|u\| \leq 1, \quad S = \mathrm{diag}(s_t I_3,\, s_r I_3). \tag{2}$$

$T(x, y)$ is the minimum time to steer x to y. Obstacle-free experiments use $s_t = s_r = 1$, the box $t \in [-1, 1]^3$, $L_{\mathrm{ref}} = 1$ and $\tau_{\mathrm{ref}} = L_{\mathrm{ref}}/s_t = 1$.

2.2 Intrinsic endpoint derivatives and the Eikonal equations

Let $E_i = Se_i$ be the scaled body twists ($i \leq 3$ translations, $i \geq 4$ rotations). The scaled tangent derivatives at the two endpoints are

$$c_x^{(i)} = \frac{d}{d\epsilon} T(x \exp(\epsilon \hat{E}_i), y)\Big|_{\epsilon=0}, \qquad c_y^{(i)} = \frac{d}{d\epsilon} T(x, y \exp(\epsilon \hat{E}_i))\Big|_{\epsilon=0}. \tag{3}$$

Dynamic programming over a short interval gives $T(x, y) = \min_{\|u\|\leq 1}[dt + T(x \exp(dt\, \widehat{Su}), y)]$, hence $0 = \min_{\|u\|\leq 1}(1 + c_x^\top u)$, and since the control set is symmetric the same holds at the goal:

$$\|c_x\| = 1, \qquad \|c_y\| = 1 \qquad \text{(away from the singular sets)}. \tag{4}$$

The specification constrains the endpoints separately; a joint condition $\|c_x\|^2 + \|c_y\|^2 = 2$ alone would let the two terms compensate.

Proposition 1 (Tangent projection)**.** *If $T(x, y) = F(G(x), G(y))$ with F differentiable, $g_t = \partial F/\partial t_x$ and $G_R = \partial F/\partial R_x \in \mathbb{R}^{3\times 3}$, then*

$$c_x = \Big(s_t\, R_x^\top g_t,\quad s_r\,(A - A^\top)^\vee\Big), \qquad A = R_x^\top G_R, \tag{5}$$

where $(\cdot)^\vee$ extracts the axial vector of a skew matrix; analogously for c_y.

Proof. For $i \leq 3$, $x \exp(\epsilon \hat{E}_i) = (t_x + \epsilon s_t R_x e_i, R_x)$, so $c_x^{(i)} = s_t\, g_t^\top R_x e_i$. For a rotation, $x \exp(\epsilon \hat{E}_{3+j}) = (t_x, R_x \exp(\epsilon s_r [e_j]_\times))$ and $c_x^{(3+j)} = s_r \langle G_R, R_x [e_j]_\times \rangle_F = s_r \langle A, [e_j]_\times \rangle_F = s_r\, e_j^\top (A - A^\top)^\vee$, using $\langle M, [w]_\times \rangle_F = w^\top (M - M^\top)^\vee$. □

Training uses (5): one autograd call with respect to G followed by the projection, so only physical tangent directions are differentiated, never redundant matrix entries as independent motions. A second implementation differentiates through $x\exp(\widehat{S\xi})$ at $\xi = 0$ with a series-safe exponential; both agree with central finite differences (Section 5.1).

2.3 Analytic solution, latent target and decoder

For this speed model the minimum time is the weighted product-metric geodesic distance: translation moves on a straight line and rotation along a shortest geodesic of SO(3),

$$T^*(x,y) = \sqrt{\frac{\|t_x - t_y\|^2}{s_t^2} + \frac{\theta^2}{s_r^2}}, \qquad \theta = \angle(R_x^\top R_y) \in [0,\pi]. \tag{6}$$

The specification's latent target and decoder are

$$a^* = \|t_x - t_y\|^2, \qquad b^* = \frac{3 - \operatorname{tr}(R_x^\top R_y)}{4}, \tag{7}$$

$$\phi(b) = 4\arcsin^2\sqrt{b}, \qquad D(a,b) = \sqrt{\frac{a}{s_t^2} + \frac{\phi(b)}{s_r^2}}. \tag{8}$$

Since $\operatorname{tr} R = 1 + 2\cos\theta$,

$$b^* = \frac{1-\cos\theta}{2} = \sin^2\frac{\theta}{2} = \frac{\|R_x - R_y\|_F^2}{8} = 1 - (q_x \cdot q_y)^2, \tag{9}$$

with q a unit quaternion of R, so $\phi(b^*) = \theta^2$ and $D(a^*, b^*) = T^*$. The latents are smooth (polynomial in the entries of G) across both singular sets; the decoder creates the corners. The oracle used for every label and error is computed independently of D: $\theta = \operatorname{atan2}\big(\frac{1}{2}\|(Q - Q^\top)^\vee\|,\ \frac{1}{2}(\operatorname{tr} Q - 1)\big)$ with $Q = R_x^\top R_y$.

Singularities of the decoder. The derivative

$$\phi'(b) = \frac{4\arcsin\sqrt{b}}{\sqrt{b(1-b)}} \tag{10}$$

has a removable singularity at $b = 0$ ($\phi'(0) = 4$) and a genuine one at $b \to 1$. For $b < 10^{-3}$ we use

$$\phi(b) = 4b + \tfrac{4}{3}b^2 + \tfrac{32}{45}b^3 + \tfrac{16}{35}b^4 + \tfrac{512}{1575}b^5 + O(b^6), \tag{11}$$

and the exact branch is evaluated on a safe substitute input so that the unused branch cannot inject NaN into the backward pass. Every learned range adapter keeps $b \le 1 - \varepsilon$, $\varepsilon = 10^{-6}$: at a saturated $b = 1$ the product $\phi'(1)\cdot 0$ gives NaN gradients. Exact half-turns are therefore representable only to within $2\sqrt{\varepsilon} = 2 \times 10^{-3}$ rad, as the specification anticipates.

2.4 The smooth graph

The learned map is

$$H_\theta(x,y) = \big(G(x), G(y), Z_\theta(x,y)\big) \in \mathbb{R}^{26}, \qquad T_\theta(x,y) = D\big(a_\theta(x,y), b_\theta(x,y)\big), \qquad dT_\theta = dD \circ dH_\theta. \tag{12}$$

Because G is fixed, H_θ is injective and its image is a 12-dimensional graph. A two-scalar latent does not identify (a,b) from T alone; latent errors are reported only as diagnostics.

3 Learning formulation

3.1 Losses

Regular samples. Derivative losses use only pairs that are regular in *input geometry* (never predicted latents): $\mathcal{R} = \{(x, y) : T^*(x, y)/\tau_{\text{ref}} \geq 10^{-3},\ 1 - b^*(x, y) \geq 10^{-4}\}$. Derivative-dependent decoder evaluations are constructed only on $\mathcal{R}$; value losses use separate forward batches.

Endpoint Eikonal.

$$L_E = \mathbb{E}_{(x,y)\in\mathcal{R}}\Big[(\|c_x\|^2 - 1)^2 + (\|c_y\|^2 - 1)^2\Big]. \tag{13}$$

Finite-step Bellman. With an EMA target network T^- (decay 0.995) and step h,

$$(\mathcal{B}T)(x, y) = \min_{u\in\mathcal{U}(x,y)}\Big[h + T^-\big(x\exp(h\,\widehat{Su}), y\big)\Big], \qquad L_B = \mathbb{E}\Big[\big(T_\theta(x, y) - \text{sg}[\mathcal{B}T](x, y)\big)^2\Big]/\tau_{\text{ref}}^2. \tag{14}$$

The admissible candidates $\mathcal{U}$ are the online greedy control $u_0 = -c_x/\|c_x\|$, fifteen normalised Gaussian perturbations of u_0 ($\sigma = 0.3$), sixteen uniform unit controls and $u = 0$. Terminal treatment: if the constant body twist reaching y exactly has duration $\tau_{\text{reach}}(x, y) = \|S^{-1}\log(x^{-1}y)\| \leq h$, the target is $\min(\mathcal{B}T, \tau_{\text{reach}})$, i.e. only the elapsed time is charged. This uses pose geometry only.

Goal, anchors and supervision.

$$L_{\text{goal}} = \mathbb{E}_x\big[Q_\theta(x, x)\big]/\tau_{\text{ref}}^2, \qquad Q_\theta = a_\theta/s_t^2 + \phi(b_\theta)/s_r^2 = T_\theta^2, \tag{15}$$

$$L_T = \frac{1}{|\mathcal{A}|}\sum_{(x,y)\in\mathcal{A}}\big(T_\theta(x, y) - T^*(x, y)\big)^2/\tau_{\text{ref}}^2, \qquad |\mathcal{A}| = 1024 \text{ declared anchors}, \tag{16}$$

$$L_Z = \mathbb{E}\big[(a_\theta - a^*)^2/L_{\text{ref}}^4 + (b_\theta - b^*)^2\big] \quad \text{(stage B only)}. \tag{17}$$

L_{goal} is computed through Q so the square root is never differentiated at zero. The physics objective is the specification's

$$L = L_E + L_B + 10\,L_{\text{goal}} + L_T. \tag{18}$$

Oracle labels enter physics training only through the declared anchor set $\mathcal{A}$; a unit test checks that the loss calls the oracle only for the regular mask.

3.2 Data

Translations are uniform in $[-1, 1]^3$; rotations are Haar-uniform (normalised Gaussian quaternions). A training batch is a declared mixture: 60 % uniform pairs, 20 % small relative perturbations $y = x\exp(r\,\widehat{Su})$ with r log-uniform in $[10^{-3}, 0.3]$, and 20 % near-half-turn pairs $R_y = R_x\exp((\pi - \delta)[n]_\times)$ with δ log-uniform in $[10^{-3}, 0.5]$; a separate diagonal batch feeds L_{goal}. Held-out sets are generated from dedicated seeds: validation (seed 1001, checkpoint selection) and test (seed 2002, all reported numbers), each with 10^4 uniform and 10^4 mixture pairs plus singularity suites of 10^3 pairs (exact half-turns, near half-turns with $\delta \in [10^{-6}, 10^{-1}]$, near-diagonal pairs with $r \in [10^{-5}, 10^{-2}]$, pure translations, pure rotations, diagonal). Training seeds are 0, 1, 2 and the anchor seed is 777.

3.3 Network: the G-Shape encoder

All heads share the encoder of `models/metric/model_network_metric.py`. With the Fourier matrix $B \in \mathbb{R}^{128\times d}$, $B_{jk} = \sigma_k \zeta_{jk}$, $\zeta \sim \mathcal{N}_{[-2,2]}(0,1)$,

$$\gamma(g) = \big[\sin(2\pi Bg);\ \cos(2\pi Bg)\big] \in \mathbb{R}^{256}. \tag{19}$$

Linear layers are Lipschitz row-normalised, $N(W)_{i:} = W_{i:}(1 + 10^{-5} - \mathrm{softplus}_{10}(1 - 1/\|W_{i:}\|))$ (norm treated as a constant). With $U = \sin(N(W_U)\gamma + \beta_U)$, $V = \sin(N(W_V)\gamma + \beta_V)$ and $z_0 = \gamma$, each of two gated residual blocks applies twice

$$z \leftarrow U \odot s + V \odot (1 - s), \qquad s = \sin\big(N(W)z + \beta\big), \tag{20}$$

and then $z_\ell = (1 - \alpha_\ell) z_{\ell-1} + \alpha_\ell \sin(N(W_{\ell,3})z + \beta_{\ell,3})$ with $\alpha_\ell = \mathrm{sigmoid}(0.1 w_\ell)$ initialised at $1/2$. The embedding is $e = N(W_o)z_2 + \beta_o \in \mathbb{R}^{256}$, optionally standardised per sample. The *input skip* appends the fixed embedding, $e \leftarrow [e; G(x)] \in \mathbb{R}^{268}$. Both endpoints pass through the same encoder (siamese) in one batch. The encoder has about 6×10^5 parameters; every head adds less than 5 %.

3.4 Pair heads

`graph_gram` (proposed). With $L(e) \in \mathbb{R}^{4\times 4}$ an affine function of e,

$$a = \|A(e_x - e_y)\|^2, \qquad P = \frac{LL^\top}{\|LL^\top\|_F}, \qquad b = (1-\varepsilon)\,\tfrac{1}{2}\|P_x - P_y\|_F^2 = (1-\varepsilon)(1 - \langle P_x, P_y\rangle_F), \tag{21}$$

$A \in \mathbb{R}^{16\times 268}$.

Proposition 2 (Properties of `graph_gram`)**.** *(i) $T_\theta(x,y) = T_\theta(y,x)$. (ii) $a(x,x) = b(x,x) = 0$, hence $T_\theta(x,x) = 0$ exactly. (iii) $b \in [0, 1-\varepsilon]$. (iv) With the input skip, the analytic latents are representable: A selecting t gives $a = a^*$, and $L = qq^\top$ gives $P = qq^\top$ and $b = (1-\varepsilon)b^*$.*

Proof. (i)–(ii) are immediate. (iii) P_x, P_y are positive semidefinite with unit Frobenius norm, so $0 \le \langle P_x, P_y\rangle_F \le 1$. (iv) $qq^\top$ is invariant to $q \mapsto -q$ and affine in R (e.g. $q_w^2 = \frac{1+\mathrm{tr}\,R}{4}$, $q_w q_x = \frac{R_{32}-R_{23}}{4}$, $q_x q_y = \frac{R_{12}+R_{21}}{4}$), so it is an affine function of $G(x)$; with $L = qq^\top$, $LL^\top = qq^\top$ has unit norm and $\langle q_x q_x^\top, q_y q_y^\top\rangle_F = (q_x \cdot q_y)^2$, and (9) gives the claim. □

Remark 1 (Why not a unit-vector embedding)**.** *A head $b = \frac{1}{2}(1 - u_x \cdot u_y)$ with unit vectors $u(R)$ would need the kernel $\cos\theta(R,R')$ to be positive semidefinite on $\mathrm{SO}(3)$. It is not: for the identity and the half-turns about the three coordinate axes all pairwise relative angles equal π, the Gram matrix is $2I - \mathbf{1}\mathbf{1}^\top$ and has eigenvalue -2. Using quaternions directly fails for the dual reason: no continuous choice of sign $q(R)$ exists on $\mathrm{SO}(3)$. The PSD Gram construction avoids both.*

Comparison heads.

- `graph_sqdist`: $b = (1-\varepsilon)\,\mathrm{smin}_\delta\big(\|B_p(e_x - e_y)\|^2\big)$ with $\mathrm{smin}_\delta(z) = z - \delta\,\mathrm{softplus}((z-1)/\delta) + \delta\log(1 + e^{-1/\delta})$.
- `graph_joint` (the specification's optional adapter): a joint encoder on $[G(x); G(y)]$, raw outputs averaged under endpoint exchange, $a = \mathrm{softplus}(r_1)$, $b = (1-\varepsilon)\,\mathrm{sigmoid}(r_2)$.

- `metric` (G-Shape baseline head): on per-sample standardised embeddings $\hat{e}$,

$$m(x,y) = \kappa \sum_{g=1}^{16} \frac{1}{10}\Big(\log \sum_{k=1}^{16} e^{10\sqrt{(\hat{e}_{x,gk}-\hat{e}_{y,gk})^2+10^{-6}}} - \log 16\Big), \qquad \kappa = 0.2, \tag{22}$$

  and $T = m$: a sum of 16 smoothed ℓ_∞ group distances, i.e. a learned metric.

- `scalar`: $T = \mathrm{softplus}\big(\frac{1}{2}(f([G_x; G_y]) + f([G_y; G_x]))\big)$.

- `graph_metric` (stage E hybrid): `graph_gram` with $a = (m - m_0)_+^2$, m from (22) and m_0 its value at equal embeddings, so $T(x,x) = 0$ is kept.

3.5 Optimisation, curriculum and recipe

The final obstacle-free recipe (`configs/graph_method/graph_gram.json`) uses batch 1024, 12 000 Adam updates, gradient clipping at norm 1, Bellman step $h = 0.2\,\tau_{\mathrm{ref}}$, Fourier $\sigma = 0.03$, and cosine decay $\eta_k = \eta_{\min} + \frac{1}{2}(\eta_0 - \eta_{\min})(1 + \cos(\pi k/N))$ with $\eta_0 = 10^{-3}$, $\eta_{\min} = 5 \times 10^{-5}$. The *relative-angle curriculum* admits only training pairs and anchors with

$$\theta(x,y) \le \theta_{\max}(k) = \theta_0 + (\pi - \theta_0)\min(1, k/K), \qquad \theta_0 = 60^\circ,\ K = 7000. \tag{23}$$

It uses input geometry only, like the regular mask. Checkpoints are selected on the validation set by the sum of median and 95th-percentile relative errors on its uniform and mixture parts, never on the training residual. All baselines use the same data stream, update budget, batch, encoder size, pose features and planners.

4 Evaluation methodology

Field metrics. Relative error $|T_\theta - T^*|/T^*$ on pairs with $T^* \ge 0.05\tau_{\mathrm{ref}}$ (median and 95th percentile), absolute errors on the singularity suites, the endpoint residual $\big|\|c_x\|^2 - 1\big|$ on regular pairs, symmetry $|T(x,y) - T(y,x)|$, goal value $T(x,x)$, the fraction of 10^4 random triples with $T(x,y) - T(x,z) - T(z,y) > 10^{-4}$ (triangle violations), and errors stratified by relative angle (0–90°, 90–150°, 150–170°, 170–179°, 179–180°). All errors are measured against the float64 atan2 oracle.

Planners. Both act on SE(3) with $x_{k+1} = x_k \exp(h_k\, \widehat{Su_k})$, $h = 0.02\tau_{\mathrm{ref}}$, at most 600 steps, and the same terminal action (jump to y when $\tau_{\mathrm{reach}}(x_k, y) \le h$, charging τ_{reach}). The *gradient rollout* uses $u_k = -c_x/\|c_x\|$ of the decoded time and halves h_k up to four times until T decreases; otherwise, or for non-finite gradients, it takes one sampling step and counts a fallback. The *sampling rollout* evaluates 128 normalised candidates per step (zero action, previous action, 63 perturbations of it, 63 uniform) and chooses $\arg\min_u\ h + T_\theta(x\exp(h\widehat{Su}), y)$. Success means reaching the terminal condition; the cost ratio is (elapsed time)/T^* on 1000 test pairs. With the exact field the sampling planner itself costs 1.3 % (cost ratio 1.013) because of its finite candidate set.

5 Development history: what was tried

About 185 training runs were made on four RTX 3090 GPUs; `EXPERIMENTS.md` logs all of them. This section summarises the steps that changed the outcome.

Table 1: Stage B, supervised fit ($L_Z + L_T + 10L_{\text{goal}}$, 2000 steps, batch 256), validation median relative error / mixture 95th percentile (%) vs. Fourier σ. The mixture contains small relative poses; heads without an exact diagonal zero fail there.

head	$\sigma = 0.03$	0.08	0.15	0.25
`graph_gram`	0.62 / 1.9	0.95 / 2.9	1.68 / 6.0	8.33 / 43.7
`graph_sqdist`	1.60 / 6.6	2.30 / 9.8	5.53 / 18.2	14.25 / 89.2
`graph_joint`	1.46 / 96.8	2.61 / 92.8	10.45 / 79.3	44.96 / 92.5
scalar	1.67 / 97.3	2.34 / 96.0	9.92 / 93.6	56.43 / 96.0
metric head	5.56 / 37.7	4.53 / 29.0	4.95 / 24.2	8.57 / 37.0

5.1 Stage A verification

All 19 unit tests pass in float64: SO(3)/SE(3) exp–log round trips including near π, orthogonality and determinant, quaternion-sign invariance, symmetry and left-invariance of the oracle, pure translation/rotation times, decoder vs. independent oracle ($< 10^{-8}$ relative), series and derivative of ϕ, tangent projection = exp-perturbation autodiff = central finite differences (with a step sweep), both Eikonal equations ($< 10^{-6}$), Bellman admissibility of the oracle, absence of oracle-label leakage, reproducible sampling and full rollouts. The analytic field evaluated with the full pipeline has relative error $\leq 6.7 \times 10^{-16}$, exact half-turn error 8.6×10^{-9}, Eikonal residual $\leq 2.8 \times 10^{-14}$, and planning cost ratio 1.00001 (gradient) / 1.012 (sampling). Two bugs were caught: a 6×10^{-12} series truncation error (switch lowered to $b < 10^{-3}$) and NaN gradients from saturated adapters ($b \leq 1 - \varepsilon$).

5.2 Fourier frequency for pose features

In the first runs every head failed, even with oracle labels; the joint networks predicted $T \approx 0$. A Fourier row B_j with i.i.d. entries of standard deviation σ has $\|B_j\| \approx \sigma\sqrt{d}$ cycles per unit, i.e. 1.7–2.4 cycles per unit for $d = 12$ or 24 at the G-Shape value $\sigma = 0.5$ (tuned for $d = 3$), far above the content of the quadratic latents. Table 1 shows the stage-B (supervised) sweep; $\sigma = 0.03$ was used for all later obstacle-free runs. The metric head is limited to about 5 % median error even with labels: a sum of ℓ_∞ group norms cannot represent the Euclidean local norm or the cut-locus corner.

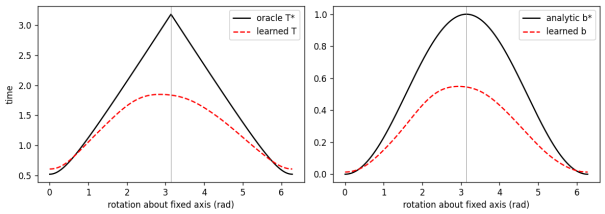
5.3 First physics runs: the cut-locus failure mode

Physics training (spec objective, 1024 anchors) plateaued at 3–5 % median and 15–20 % 95th-percentile error for the graph heads; the metric head reached 9–12 % and degraded with longer training. Recipe changes (Bellman step 0.02–0.5, Bellman weight, EMA decay, learning rate, anchor weight, batch 4096, residual forms $(\|c\| - 1)^2$ and $\log^2 \|c\|^2$, T-weighted residuals, Gram sharpening $P \propto (LL^\top)^k$, narrower cut-locus band, more near-π samples) moved the plateau by less than one point or made it worse. Removing the Bellman term changed little.

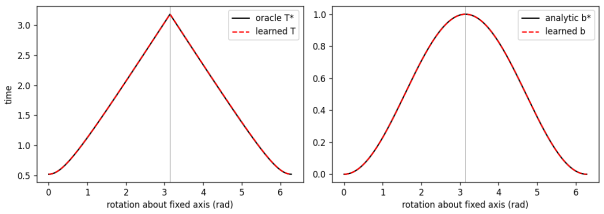
Table 2 and Figure 1(a) show the mechanism. Near the half-turn the learned b stays below one with a smooth maximum, so its slope, and with it the rotational gradient, vanish. The network then satisfies $\|c_x\| \approx 1$ with spurious translation sensitivity. A smooth saturating adapter makes this worse: for `graph_sqdist` with $\delta = 0.02$, $dT/dz \propto \phi'(b)\, db/dz \approx 2\pi\sqrt{1-b}/\delta \to 0$ as $b \to 1$.

Table 2: Diagnosis on near-pure rotations ($\|\Delta t\| = 0.3$) of plateaued checkpoints: squared norms of the translation and rotation parts of c_x (true values in parentheses), ratio T/T^* and the learned rotation latent b vs. b^*.

	`graph_sqdist`				metric head	
angle	$\|c_{x,t}\|^2$	$\|c_{x,\omega}\|^2$	T/T^*	b (b^*)	$\|c_{x,\omega}\|^2$	T/T^*
0–15°	0.76 (0.75)	0.33 (0.25)	1.03	0.011 (0.008)	0.20	1.05
90–105°	0.10 (0.03)	0.94 (0.97)	1.05	0.53 (0.57)	0.23	0.73
150–165°	0.34 (0.01)	0.42 (0.99)	0.97	0.89 (0.96)	0.09	0.58
165–180°	0.40 (0.01)	0.08 (0.99)	0.93	0.92 (0.99)	0.06	0.54



(a) early physics run (3k steps): b peaks at ≈ 0.55



(b) final `graph_gram`: b reaches 1, corner reproduced

Figure 1: Travel time (left of each panel) and rotation latent (right) along a slice that rotates the goal about a fixed axis through the cut locus at π; black: oracle, red dashed: learned.

5.4 The correct solution is a stable minimum: warm-start test

A stage-B checkpoint (0.11 % / 0.33 %) used to initialise physics training stayed in the correct basin: 1.8 % / 5.1 % at learning rate 10^{-3} with anchors, and 0.7 % / 1.9 % at 10^{-4} even *without* anchors (warm-start runs are labelled separately and are not results). Hence training from scratch failed by reaching a wrong basin, not because the loss is biased, and gradient noise at 10^{-3} costs about two points.

5.5 Curriculum, cosine decay and input skip

Restricting the training pairs to relative angles below $\theta_{\max}(k)$ lets the siamese embedding, which is shared by all pairs, learn the right per-pose structure before the cut locus is seen. With the input skip and cosine decay, `graph_gram` reached 0.14 % / 0.41 % while still trained only on pairs up to 160°, i.e. it extrapolated almost exactly to the whole of SO(3). Without the skip the same curriculum reached only 17 % at 4k steps; without the curriculum the skip runs stayed at 4–6 %; the metric head did not benefit (16 %).

5.6 Re-entry of the cut locus and the band sweep

When the last near-π pairs enter L_E, $\phi'(b) \to \infty$ makes $(\|c\|^2 - 1)^2$ spike on a few samples. `graph_sqdist` did not recover; `graph_gram` dropped to 1.7 % / 4.8 % and recovered under the decaying learning rate to 0.10 % / 0.32 %. A cut-locus band of 10^{-3} softens the transient; 10^{-2} removes too much supervision (0.38 % / 0.69 %).

Table 3: Field accuracy. Relative errors in %; "mix" is the training mixture (includes small relative poses and near half-turns); exact-π is the mean absolute error on exact half-turns ($T^* \approx 3.4$); Eikonal residual is the median $|\|c_x\|^2 - 1|$ on regular pairs.

model	seeds	median	95th pct.	mix 95th	179–180° median	exact-π abs.	Eikonal res.
`graph_gram` (proposed)	3	0.1050 ± 0.0029	0.313 ± 0.023	0.341 ± 0.028	0.0499 ± 0.0040	0.0021 ± 0.0003	0.0041 ± 0.0005
`graph_sqdist`	3	0.405 ± 0.051	2.56 ± 0.21	3.84 ± 0.50	1.1 ± 1.4	0.068 ± 0.023	0.0245 ± 0.0020
metric, best recipe	3	9.53 ± 0.15	23.09 ± 0.14	24.09 ± 0.16	12.87 ± 0.21	0.4553 ± 0.0099	0.1391 ± 0.0025
metric, our recipe	3	12.40 ± 0.15	35.06 ± 0.32	38.00 ± 0.24	26.63 ± 0.42	0.9097 ± 0.0063	0.0880 ± 0.0010
metric, NTFields loss	3	10.50 ± 0.11	33.3 ± 1.3	31.44 ± 0.57	12.86 ± 0.61	0.453 ± 0.019	0.3965 ± 0.0015
scalar T	3	5.70 ± 0.51	22.4 ± 1.3	40.6 ± 2.7	14.8 ± 1.5	0.520 ± 0.048	0.130 ± 0.017

Table 4: Planning on 1000 test pairs (identical planners for every field) and metric properties. Cost ratio = path time / T^*. Goal values $T(x, x)$ are exactly 0 for the graph heads, 0.003 for the metric heads (their smoothing offset) and 0.057 for the scalar network; among 10^4 random triples no model violates the triangle inequality by more than 10^{-4} except the NTFields-loss metric head (0.003 % of triples).

		gradient rollout			sampling rollout	
model	seeds	success %	cost median	cost 95th	success %	cost median
`graph_gram` (proposed)	3	100.0 ± 0.0	1.0000 ± 0.0000	1.0001 ± 0.0000	100.0 ± 0.0	1.0127 ± 0.0000
`graph_sqdist`	3	100.0 ± 0.0	1.0001 ± 0.0000	1.0024 ± 0.0010	100.0 ± 0.0	1.0130 ± 0.0001
metric, best recipe	3	98.63 ± 0.21	1.1179 ± 0.0027	1.381 ± 0.026	98.57 ± 0.35	1.1367 ± 0.0013
metric, our recipe	3	95.00 ± 0.61	1.3405 ± 0.0072	1.937 ± 0.036	97.47 ± 0.15	1.2298 ± 0.0031
metric, NTFields loss	3	92.43 ± 0.87	1.1280 ± 0.0034	1.586 ± 0.033	92.43 ± 0.55	1.1487 ± 0.0036
scalar T	3	1.43 ± 0.61	1.3893 ± 0.0040	2.31 ± 0.77	1.93 ± 0.61	1.216 ± 0.040

5.7 A degeneracy of physics-only training

Without anchors, rotations were learned well (pure rotations 0.9 % / 2.7 %) but translations were not (9.7 % / 66 %), with $T/\|\Delta t\|$ down to 0.35 even for tiny displacements.

Proposition 3 (Projection-metric degeneracy)**.** *Suppose near the diagonal* $T(x, x\exp(\widehat{S\delta})) = \sqrt{\delta^\top M \delta} + o(\|\delta\|)$ *with* $M \succeq 0$*. The endpoint Eikonal equation holds for all small* δ *if and only if* $M^2 = M$*, i.e.* M *is any orthogonal projection, not only* $M = I$*.*

Proof. To first order $c_x = -M\delta/\sqrt{\delta^\top M \delta}$, so $\|c_x\|^2 = \delta^\top M^2 \delta / \delta^\top M \delta$. This equals one for all δ iff the symmetric matrix $M^2 - M$ vanishes. □

Globally, in a flat analogue $T = \|P(x - y)\|$ with a projection P has $\|\nabla_x T\| = 1$ wherever it is differentiable and spurious zeros on $x + \ker P$. The Bellman minimum selects the steepest action and never charges a zero-cost direction, so it only lifts the spurious zero set itself, by about h per target update. Declared anchors remove the degeneracy; without labels it is the dominant failure. A non-detached $e^{-T/2}$ residual weight did not help here.

6 Results: obstacle-free SE(3) (stages A–D)

Tables 3–6 report the held-out test set (seed 2002) as mean ± standard deviation over training seeds 0, 1, 2. Rows marked with one seed are single runs.

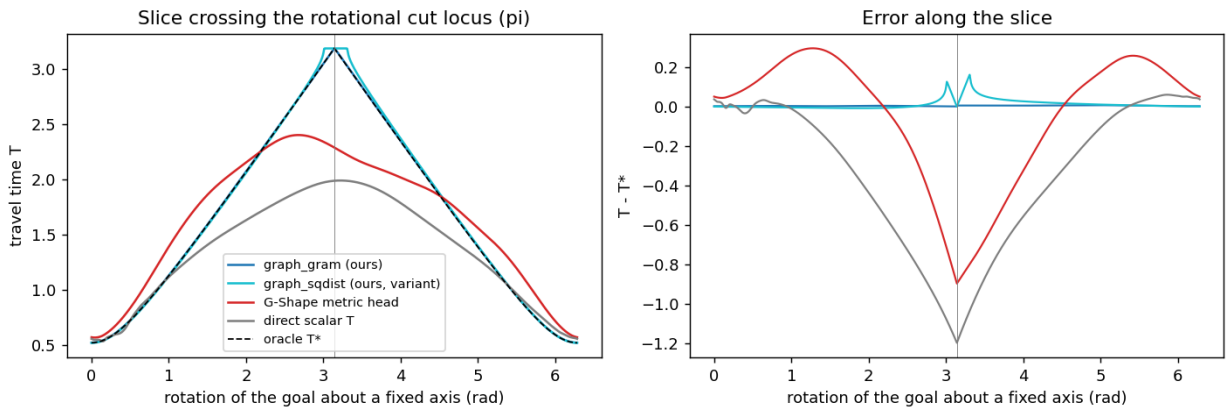


Figure 2: Seed-0 fields along the cut-locus slice of Figure 1. Only the graph heads form the corner at π; the metric head and the scalar network are 0.9 and 1.2 too low there. `graph_sqdist` shows a short flat top caused by its saturating adapter.

Table 5: Median relative error (%) by relative rotation angle (Figure 3).

model	0–90°	90–150°	150–170°	170–179°	179–180°
`graph_gram` (proposed)	0.180 ± 0.014	0.1096 ± 0.0033	0.0839 ± 0.0017	0.0717 ± 0.0010	0.0499 ± 0.0040
metric, best recipe	15.80 ± 0.32	8.17 ± 0.16	6.033 ± 0.079	11.15 ± 0.28	12.87 ± 0.21
scalar T	5.72 ± 0.81	4.73 ± 0.35	7.8 ± 1.5	12.8 ± 1.7	14.8 ± 1.5

Findings (empirical, this benchmark).

- The smooth-graph field is about 90× more accurate than the G-Shape metric head under identical conditions and plans optimally (cost ratio 1.0000 vs. 1.12). It meets the specification's provisional gates on every seed.
- The mechanism is confirmed at the cut locus. The error of `graph_gram` *decreases* toward 180°, the decoded time has the corner (slice error ≤ 0.0041 near π), and the latents match the analytic ones without latent supervision (median $|a - a^*| = 0.013$, $|b - b^*| = 4.3 \times 10^{-4}$). This identifiability was observed, not guaranteed.
- Exact goal values matter for planning. The direct scalar network cannot be planned on with the gradient rollout (1.4 % success): without an exact zero its minimum is offset from the goal.
- Required ingredients: the curriculum (without: 4.0 %), the input skip (without: 11.8 %) and the anchors (without: 2.4 % / 19 %, Proposition 3).
- The Bellman term, as configured, slightly *hurts* final accuracy (0.072 % without vs. 0.105 % with). It is kept in the main row because the specification's objective includes it. Finite candidate sets bias $\mathcal{B}T$ upward, and the lagged EMA target slows convergence.

7 Stage E: obstacles

The specification allows this only after stages A–D and warns that the two-output decoder then no longer implies smooth targets or physical meanings for a and b. Everything in this section is empirical.

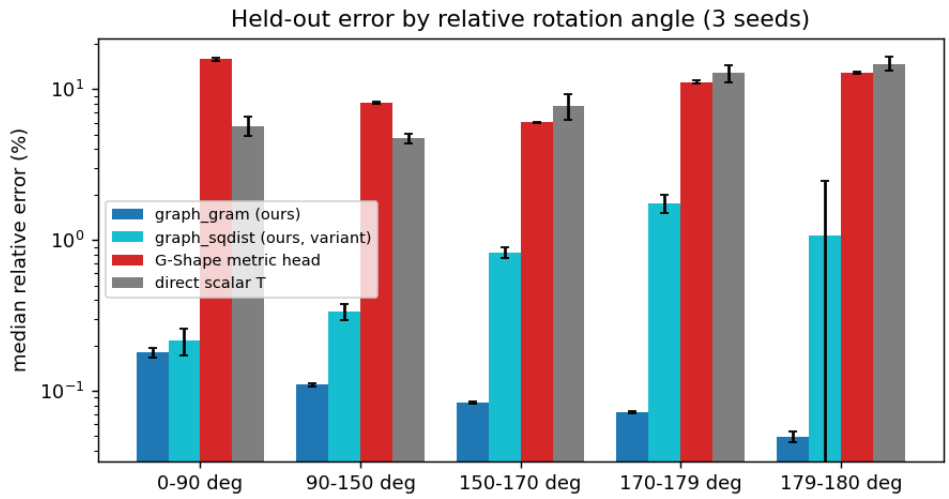


Figure 3: Median relative error by relative rotation angle (log scale, 3 seeds).

7.1 Setup

Data and speed model. The preprocessed 3D-shape datasets of the repository (in `datasets/3dshape/`, 8×10^5 pose pairs each, rotations stored as rotation vectors divided by 2π) provide clearance speeds $s \in (0, 1]$ per endpoint. We use the declared position-dependent ellipsoidal model

$$\frac{\|v\|^2}{(S(x)s_t)^2} + \frac{\|\omega\|^2}{(S(x)s_r)^2} \le 1, \qquad s_t = 1,\ s_r = 2\pi, \qquad S = w(s) = s^2(2-s)^2, \tag{24}$$

i.e. rotations are measured in turns as in the baseline, and w is the baseline's speed warp applied to the combined (isotropic) speed. The endpoint Eikonal equations become $\|S(x)\, c_x\| = \|S(y)\, c_y\| = 1$ with c computed by (5).

Loss. No oracle and no anchors exist. Following NTFields we use the residual $r_x = (\|S(x)c_x\|^{1/2} - 1)^2$ and the weight $e^{-T/2}$ *without* detaching T:

$$L = \mathbb{E}\big[(r_x + r_y)\, e^{-T_\theta/2}\big], \qquad \nabla_\theta L = \mathbb{E}\big[e^{-T_\theta/2}\nabla_\theta(r_x + r_y)\big] - \tfrac{1}{2}\,\mathbb{E}\big[(r_x + r_y)\, e^{-T_\theta/2}\,\nabla_\theta T_\theta\big]. \tag{25}$$

The second term pushes T up wherever the residual is large, an implicit anti-collapse force (Proposition 3). There is no Bellman term (the baseline runs with TD off). AdamW (weight decay 0.5, learning rate 5×10^{-4}), batch 2000, 20 000 updates, Fourier $\sigma_t = 0.5$ for translation and $\sigma_r = 0.1$ for rotation columns (`configs/graph_method/stageE_common.json`). All heads use exactly this pipeline.

Evaluation. The unmodified Shape-NTFields evaluator (`evaluate_training_3d_batched.py`: MPPI in $(t, \text{rotvec}/2\pi)$ coordinates, collision checks against the environment mesh, 1000 RRT-verified start/goal pairs) queries our fields through a small adapter (`shape_adapter.py`). It reports four planner variants (forward, reverse, either-of-the-two, alternate) and failure counts (collision, non-convergence; they can overlap).

Table 6: Ablations of the `graph_gram` recipe, the stage-B supervised reference and the stage-E hybrid head on the obstacle-free benchmark. Columns as in Tables 3 and 4.

variant	seeds	median	95th pct.	mix 95th	179–180° median	exact-π abs.	Eikonal res.
`graph_gram` (full)	3	0.1050 ± 0.0029	0.313 ± 0.023	0.341 ± 0.028	0.0499 ± 0.0040	0.0021 ± 0.0003	0.0041 ± 0.0005
w/o curriculum	3	4.01 ± 0.34	19.2 ± 1.1	21.9 ± 1.1	13.3 ± 1.1	0.482 ± 0.039	0.0871 ± 0.0073
w/o anchors	3	2.42 ± 0.41	19.03 ± 0.33	18.22 ± 0.24	2.15 ± 0.46	0.128 ± 0.013	0.031 ± 0.010
w/o input skip	3	11.79 ± 0.17	36.30 ± 0.61	38.66 ± 0.92	27.26 ± 0.13	0.9334 ± 0.0071	0.1169 ± 0.0004
w/o Bellman	3	0.0716 ± 0.0080	0.239 ± 0.028	0.275 ± 0.038	0.0688 ± 0.0070	0.0037 ± 0.0002	0.0040 ± 0.0006
band 10^{-3}	3	0.165 ± 0.060	0.360 ± 0.052	0.377 ± 0.044	0.100 ± 0.035	0.0036 ± 0.0025	0.0033 ± 0.0007
stage B (labels)	1	0.026	0.08	0.10	0.07	0.0040	0.0016
`graph_metric`	1	8.724	29.13	32.23	21.99	0.7605	0.0753

variant	seeds	grad. succ. %	cost med.	cost 95th	samp. succ. %	cost med.
`graph_gram` (full)	3	100.0 ± 0.0	1.0000 ± 0.0000	1.0001 ± 0.0000	100.0 ± 0.0	1.0127 ± 0.0000
w/o curriculum	3	100.0 ± 0.0	1.0431 ± 0.0045	1.2378 ± 0.0066	100.0 ± 0.0	1.0583 ± 0.0050
w/o anchors	3	100.0 ± 0.0	1.65 ± 0.24	2.16 ± 0.43	99.73 ± 0.31	1.250 ± 0.037
w/o input skip	3	76 ± 15	1.124 ± 0.017	1.513 ± 0.015	76 ± 15	1.143 ± 0.017
w/o Bellman	3	100.0 ± 0.0	1.0000 ± 0.0000	1.0001 ± 0.0000	100.0 ± 0.0	1.0125 ± 0.0001
band 10^{-3}	3	100.0 ± 0.0	1.0000 ± 0.0000	1.0001 ± 0.0000	100.0 ± 0.0	1.0126 ± 0.0001
stage B (labels)	1	100.0	1.0000	1.000	100.0	1.013
`graph_metric`	1	99.1	1.1455	1.400	99.2	1.160

7.2 Getting the pipeline to work

Table 7 lists the exploration on the first 200 test cases (one seed each; standard error about 3 points). The first fields failed completely (0–1 %; every episode failed to converge). Two diagnostics located the problem. (i) Since $S \le 1$, the obstacle-free time is a lower bound, $T \ge T_{\text{free}}$; the working baseline field has $T/T_{\text{free}} \approx 3.0$ (5th–95th percentile 2.4–4.1) on test pairs, ours had 0.55–1.3. (ii) Binned by clearance speed, the baseline's $S\|\nabla T\|$ has medians $0.13, 0.63, 1.08, 1.44$ for $S \in [0, 0.01), [0.01, 0.1), [0.1, 0.5), [0.5, 1)$; ours had $0.025, 0.24, 1.09, 2.0$: a nearly speed-independent gradient that ignores the clearance structure. With $c = S\nabla T$ the spec residual's push on $\|\nabla T\|$ scales like S^2 ($\approx 3 \times 10^{-6}$ in the clearance band). The non-detached weight (0–1 % $\to$ 22 %), the input skip (6 % $\to$ 22 %) and above all the rotation frequency (with $G(x)$ containing $\operatorname{vec} R$, $\sigma_r = 0.3$ gives about 3× the phase per radian of the baseline's rotation-vector input; $\sigma_r = 0.1$: 22 % $\to$ 67–69 %; then $\sigma_t = 0.5$: 77–79 %) fixed it. Anisotropic speeds hurt (11 %); a hinge penalty for $T \ge T_{\text{free}}$ did not help.

7.3 The translation-type latent: Euclidean vs. metric

With the matched pipeline the metric head led `graph_gram` by about five points. For `graph_gram`, $T^2 - \phi(b)/s_r^2 = \|A(e_x - e_y)\|^2$ is a squared Euclidean embedding distance, while the rotation term is bounded:

$$0 \le \frac{\phi(b)}{s_r^2} \le \frac{\pi^2}{(2\pi)^2} = \frac{1}{4}. \tag{26}$$

All extra time caused by obstacles must therefore be expressed as a Euclidean distance between embeddings. A finite metric is isometrically ℓ_2-embeddable only if $-\frac{1}{2} J D^{(2)} J \succeq 0$ (Schoenberg), which geodesic distances in non-convex domains can violate, whereas every finite metric embeds isometrically into ℓ_∞ (Fréchet), the structure of (22). The hybrid `graph_metric` keeps the graph latents and decoder but takes a from the metric distance. Widening the Euclidean latent instead ($k_a = 128$) did not help. The hybrid is not a universal head: on the obstacle-free benchmark it

Table 7: Stage E exploration on Tshape3d_env1, first 200 test cases, forward MPPI. σ column: Fourier σ_t/σ_r (single value: coordinate inputs). The last block calibrates against a replica of the baseline pipeline and the baseline checkpoint itself.

head	setting	σ	success %	collision	no-conv.
`graph_gram`	sq. residual, aniso speeds	1.0/0.3	0.0	178	200
`graph_gram`	sq. residual, iso speeds	1.0/0.3	0.0	158	200
`graph_gram`	$\sqrt{\cdot}$ residual, iso	1.0/0.3	1.0	130	198
`graph_gram`	+ non-detached $e^{-T/2}$ weight	1.0/0.3	22.5	143	77
`metric`	+ non-detached $e^{-T/2}$ weight	1.0/0.3	5.5	26	188
`graph_gram`	no input skip	1.0/0.3	6.0	124	186
`graph_gram`	+ speed lower bound $T \geq T_{\text{free}}$	1.0/0.3	22.0	145	71
`metric`	+ speed lower bound	1.0/0.3	10.1	45	177
`graph_gram`	AdamW (wd 0.5)	1.0/0.3	26.5	137	56
`graph_gram`	Adam, cosine LR	1.0/0.1	67.0	22	61
`graph_gram`	AdamW	1.0/0.1	69.0	32	43
`metric`	Adam, cosine LR	1.0/0.1	60.3	4	79
`metric`	AdamW	1.0/0.1	71.0	8	53
`graph_gram`	AdamW, aniso speeds	1.0/0.1	11.0	170	110
`graph_gram`	AdamW	2.0/0.1	52.5	64	56
`graph_gram`	AdamW	0.5/0.1	77.5	13	38
`graph_gram`	AdamW	0.5/0.05	78.5	15	31
`graph_gram`	AdamW	0.3/0.05	78.0	9	40
`metric`	AdamW	0.5/0.1	81.0	1	37
`metric`	coordinate replica of baseline	1.0	68.5	40	50
`metric`	replica + rollback, resampled B	1.0	74.5	34	39
`metric`	replica, no clipping, no mask	1.0	76.0	32	43
`metric baseline`	native pipeline, checkpoint s1	1.0	82.0	19	29

reaches only 8.7 % median error (Table 6). Because the b term is bounded by 1/4, among obstacles the hybrid is essentially the metric head plus a graph-style rotation term, so its parity shows that the graph structure costs nothing there, not that it helps.

7.4 Results

Table 8 gives the final comparison. Inside one pipeline `graph_metric` is on par with the metric head: 79.2 vs. 80.0 % forward on the T-shape (equal or better on the three other planner variants), 82.9 vs. 81.4 % on the L-shape and 81.5 vs. 81.5 % on the A-shape (mean 81.2 vs. 81.0 %). `graph_gram` is 5–11 points behind. The metric head inside our pipeline matches the native baseline within about one point on the T-shape, so the pipeline is validated. The native pipeline is still ahead there, by 1.2 (metric head), 2.0 (`graph_metric`) and 6.1 (`graph_gram`) points against its two re-scored checkpoints (more against its five-seed mean), and it fails differently (about 118 collisions vs. about 20 for ours). Research notes report 88–92 % (L) and 86–89 % (A) for it. The remaining gap was not attributable to the NTFields accept/reject rollback (it never triggered), gradient clipping, the regular-sample mask or the Fourier sampling scheme (Table 7); it remains unexplained. Lshape3d_env3 and Ashape3d_env4 were scored with the Lie-Shape-NTFields copy of the same evaluator (identical code path, byte-identical test sets), because the Shape-NTFields checkout lacks those meshes.

Table 8: Stage E, all 1000 test cases per environment, success (%) per MPPI variant and failure counts of the forward planner. †Native NTFields pipeline, checkpoints of seeds 1 and 2 (Lie-Shape-NTFields, epoch 4000) with the same evaluator and test set; its research notes report 81.9 ± 0.5 % over five seeds on this environment.

model	seeds	forward	reverse	either	alternate	collision	no-conv.
Tshape3d_env1							
metric NTFields baseline, native pipeline†	2	81.2 ± 1.3	80.0 ± 0.1	91.1 ± 0.8	88.6 ± 0.4	118 ± 6	139 ± 14
metric head, our pipeline	3	80.0 ± 0.6	80.1 ± 0.4	87.2 ± 0.2	82.1 ± 0.9	19 ± 6	188 ± 1
`graph_metric` (graph latents)	3	79.2 ± 0.2	80.8 ± 0.1	87.3 ± 0.2	82.5 ± 0.7	20 ± 6	194 ± 4
`graph_gram`	3	75.1 ± 0.4	76.2 ± 0.5	85.8 ± 0.3	80.0 ± 0.8	75 ± 10	200 ± 4
`graph_gram`, $k_a = 128$	1	74.2	74.6	84.8	80.1	84	196
Lshape3d_env3							
metric head, our pipeline	2	81.4 ± 0.4	81.2 ± 1.1	86.8 ± 0.2	83.8 ± 0.8	18 ± 4	174 ± 3
`graph_metric` (graph latents)	2	82.9 ± 0.7	83.2 ± 0.1	88.8 ± 0.1	84.5 ± 0.6	17 ± 3	160 ± 9
`graph_gram`	1	70.3	72.2	82.2	80.6	39	277
Ashape3d_env4							
metric head, our pipeline	2	81.5 ± 1.1	81.2 ± 0.4	87.3 ± 0.1	83.5 ± 0.1	18 ± 0	173 ± 8
`graph_metric` (graph latents)	2	81.5 ± 0.8	80.6 ± 1.4	87.5 ± 0.1	83.4 ± 0.6	22 ± 5	170 ± 5
`graph_gram`	1	75.6	75.3	84.9	82.3	50	211

8 Discussion

Established vs. empirical. Proven or verified to numerical precision: the identities of Section 2, the tangent projection, the Eikonal equations of the oracle, Proposition 2 and Proposition 3. Empirical, on this benchmark only: every learned result. Not established: convergence of sampled PDE training to the viscosity solution in general, identifiability of (a, b), the triangle inequality (measured, not guaranteed), any claim of superiority among obstacles.

Deviations from the specification (all tuned on validation data only).

- Batch 1024 instead of 256 (same wall-clock cost; training is kernel-launch bound) and 12k instead of 20k updates.
- Bellman step $h = 0.2\,\tau_{\text{ref}}$.
- The relative-angle curriculum and the input skip.
- Training is in float32; geometry verification is in float64.
- Stage E uses the NTFields residual form and weighting, isotropic speeds, no Bellman term and no anchors.

Limitations. The obstacle-free benchmark has an analytic solution; success validates the representation and its optimisation, not a need for learning. Physics-only training without labels does not reach the accuracy of the anchored version because of the degeneracy of Proposition 3. The curriculum uses the relative rotation angle of the pair, which is input geometry but specific to the rotational structure. Stage E has one environment with three seeds and two with two seeds, and an unexplained gap to the native pipeline.

Future work. (i) One head for both settings, e.g. $a = \|A(e_x - e_y)\|^2 + m^2$, letting the network use whichever channel fits; (ii) constraints that remove the projection-metric degeneracy so physics-only training works without anchors; (iii) Bellman targets without upward bias (more or optimised candidates, multi-step rollouts); (iv) closing the gap to the native NTFields pipeline; (v) more environments and seeds.

9 Reproducibility

All code runs inside the `pytorchserver:latest` image (PyTorch 2.2.2, CUDA 12.1) from the repository root.

```
python -m unittest tests.test_graph_method                          # stage A
python -m models.graph_method.evaluate --analytic --out results/graph_method/analytic
python train/train_graph.py --config configs/graph_method/graph_gram.json --seed 0 \
    --out_root ./Experiments/graph_method_final
python -m models.graph_method.evaluate \
    --ckpt Experiments/graph_method_final/graph_gram_s0/best.pt \
    --out results/graph_method_final/graph_gram_s0
python train/train_graph.py --config configs/graph_method/stageE_common.json --seed 0 \
    --name stageE_graph_metric_s0 --out_root ./Experiments/graph_method_stageE \
    --set fourier_sigma_t=0.5 fourier_sigma_r=0.1 arch=graph_metric skip_input=True
python models/graph_method/scripts/run_shape_eval.py /shape /workspace \
    --dataPath testing_data/3dshape/Tshape3d_env1 --no-viser --cases 0 \
    --checkpoint /workspace/Experiments/graph_method_stageE/stageE_graph_metric_s0/latest.pt \
    --out /workspace/results/graph_method_stageE/stageE_graph_metric_s0_c1000
python3 models/graph_method/report/make_tables.py                   # tables of this report
```

The package layout follows the specification's contract: `geometry/` (se3, tangents), `fields/` (oracle, decoder), `interface.py`, `model_network_graph.py` (encoder and heads), `losses/` (eikonal, bellman), `data/poses.py`, `planning/rollout.py`, `model_function_graph.py` and `model_train_graph.py` (losses, training, checkpoints with optimiser, EMA, sampler and RNG states, scales, exclusions and code revision), `evaluate.py`, `model_function_shape.py` and `shape_adapter.py` (stage E), `tests/test_graph_method.py` and `configs/graph_method/`. All tables in this report are generated from the saved evaluation files by `report/make_tables.py`.

A Hyperparameters

	obstacle-free (`graph_gram.json`)	stage E (`stageE_common.json`)
speed scales s_t, s_r	$1, 1$	$1, 2\pi$ (position-dependent $S(x)$)
encoder	256 wide, 2 gated blocks, skip	same
Fourier σ	0.03 (all columns), clipped	$\sigma_t = 0.5$, $\sigma_r = 0.1$, resampled
heads	`graph_gram`: $k_a = 16$, 4×4 Gram	`graph_gram`, `graph_metric`, metric
objective	$L_E + L_B + 10L_{\text{goal}} + L_T$	$\mathbb{E}[(r_x + r_y)e^{-T/2}]$
Eikonal residual	$(\|c\|^2 - 1)^2$	$(\|Sc\|^{1/2} - 1)^2$
anchors	1024 (seed 777)	none
Bellman	$h = 0.2$, 32 candidates, EMA 0.995	none
bands	$T^*/\tau_{\text{ref}} \geq 10^{-3}$, $1 - b^* \geq 10^{-4}$	$T_{\text{free}} \geq 10^{-3}$, $1 - b_{\text{free}} \geq 10^{-3}$
curriculum	$60^\circ \to 180^\circ$ over 7k updates	none
optimiser	Adam, cosine $10^{-3} \to 5 \times 10^{-5}$	AdamW, wd 0.5, 5×10^{-4}
batch / updates	1024 / 12 000	2000 / 20 000
gradient clipping	1.0	1.0
selection	best validation score	last checkpoint